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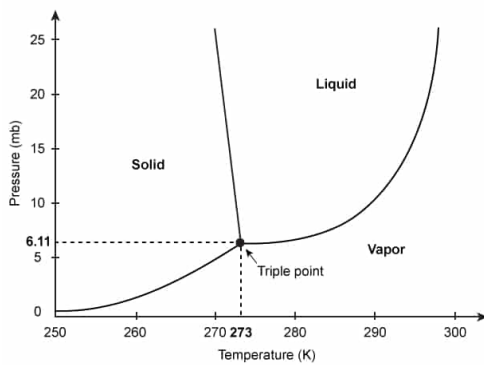


Figure 1 Phase diagram for water

In the liquid phase, water dissolves and transports molecules, and aids in the biochemical reactions necessary to support life. For this reason, scientists believe liquid-phase water is essential to the existence of extraterrestrial life. Although it has been known for some time that water is present in both the gas and ice phases on Mars, the question remains as to whether there are regions where liquid-phase water can stably exist. Estimates of Mars' surface temperature and pressure generated from satellite readings and global climate models, as well as observations made by Mars rovers, reveal that surface conditions on the planet can vary greatly.

Table 1 Temperature and Pressure of Various Locations on Mars

Location/Mission	Temperature	Pressure
Estimated planetary average	218 K	6.00 mb
Viking I	256 K	7.11 mb
Pathfinder	260 K	6.60 mb
Spirit	313 K	6.50 mb

Despite the extremely low pressures and temperatures on Mars, satellite images have repeatedly found evidence of flowing liquid water in several regions during the late Martian summer season. These regions, called recurring slope lineae (RSL), are hypothesized to be thermodynamically stable under certain Martian conditions due to the presence of dissolved salts. A simple understanding of this hypothesis is afforded by Raoult's law, which states that the vapor pressure and freezing point of an ideal solution are lowered as the number of moles of dissolved solute increases. Moreover, this effect also impacts the solution boiling point (the point at which the vapor pressure of the solution becomes equal to the ambient pressure).

The briny-water hypothesis was confirmed in 2015 when spectral data from the Compact Reconnaissance Imaging Spectrometer onboard the Mars Reconnaissance Orbiter revealed the presence of hydrated perchlorates in the same regions and times that RSL were observed. Studies are already underway to determine how common such briny water flows are and whether the conditions within them are suitable for microbial life.

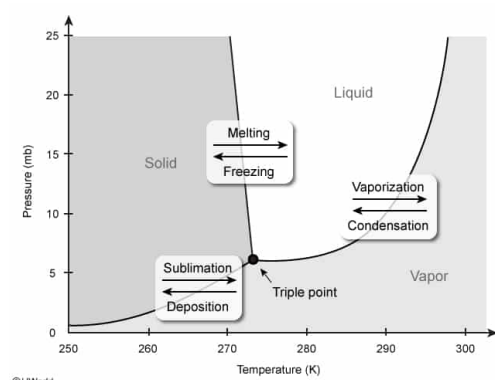
Which of the following conclusions can be made based on Figure 1 and the data in Table 1?

- ☐ A. Liquid water initially at 280 K and 15.0 mb will remain liquid at the Viking I location. [10%]
- ☐ B. A block of ice at the Pathfinder location will sublime when heated to 274 K. [16%]
- ☐ C. Water vapor released into Mars' average conditions will condense and then freeze. [12%]
- ✔ ☒ D. Liquid water initially at 274 K and 6.50 mb will boil at the Spirit location. [61%]

Correct

61% answered correctly

Explanation:



A **phase diagram** illustrates a substance's stability in a given phase (**solid, liquid, or gas**) as a function of **pressure** and **temperature**. The **triple point** is the point at which all three **boundary lines** intersect, and indicates the temperature and pressure at which all three phases simultaneously exist in equilibrium. Liquid water cannot exist at pressures below the triple point. The **boundary lines** between phases indicate the conditions under which two phases are in equilibrium. If a change in conditions results in the **crossing of a boundary line**, a substance will undergo the corresponding **change in phase**.

Liquid water initially at 274 K and 6.50 mb will cross the liquid-gas boundary line when heated to 313 K (the temperature at the Spirit location), so liquid water will boil under these conditions.

(Choice A) Conditions at the Viking I location are 256 K and 7.11 mb. Under these conditions, water enters the solid phase.

(Choice B) Sublimation can occur only at pressures *below* 6.11 mb (the pressure at the triple point of water). Because 6.60 mb (the pressure at the Pathfinder location) is greater than 6.11 mb, the ice will melt when heated to 274 K.

(Choice C) Because 6.00 mb (Mars' average pressure) is below the triple point, liquid water cannot exist under these conditions. Therefore, exposing the vapor to average Martian atmospheric conditions will cause it to undergo a phase transition directly to ice (deposition).

Educational objective:

A phase diagram shows the stability of each phase (solid, liquid, and gas) as a function of pressure and temperature. Equilibrium between two phases is indicated by phase boundaries, and crossing a phase boundary indicates a phase change. The triple point is the temperature and pressure at which all three phases exist in equilibrium.

Time spent: 0 sec

Subject:
General Chemistry

QID: 400474

System:
Thermodynamics, Kinetics & Gas Laws

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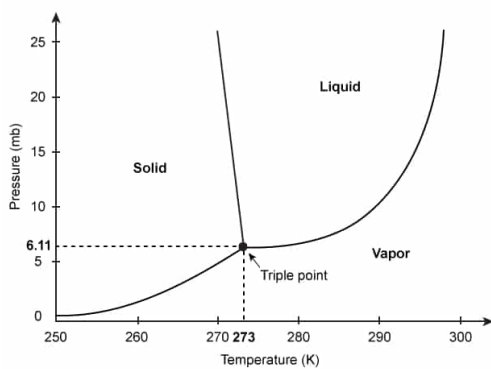


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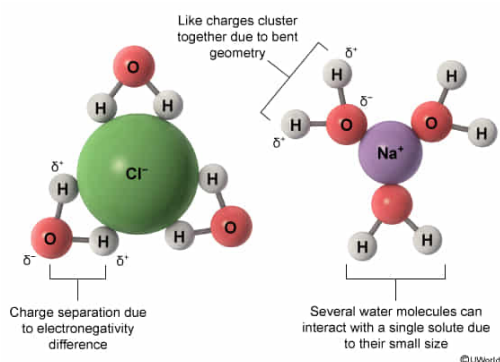
Which of the following does NOT explain water's ability to act as a solvent?

- ☐ A. Water has the geometry of a bent molecule. [6%]
- ☐ B. Hydrogen is less electronegative than oxygen.
- ☐ C. Water is a relatively small molecule. [34%]
- ✔ ☒ D. Water has a relatively high surface tension. [59%]

Correct

59% answered correctly

Explanation:



Water readily dissolves many **polar** and **ionic compounds** because it is able to interact well with them through electrostatic interactions (ie, **hydrogen bonding** and **ion-dipole interactions**). Several important characteristics contribute to water's ability to participate in these interactions, including the following:

- Because oxygen is more **electronegative** than hydrogen, the bonding electron pairs are more tightly drawn to oxygen. This property results in the oxygen atom having a partial negative charge whereas the hydrogen atoms have a partial positive charge, allowing interactions with other charged molecules (**Choice B**).
- The **bent geometry** of water contributes to its polarity by grouping positive charges at one end of the molecule and negative charges at the other. The charged regions of a water molecule are attracted to opposite charges on other polar compounds (**Choice A**).
- Water's **small size** allows efficient formation of a **hydration shell** around solutes. This process acts to evenly distribute and isolate solute particles (**Choice C**).

On the other hand, **surface tension** is a force induced at the interface between a liquid and a gas. The molecules in the liquid interact with each other more strongly than they interact with molecules in the air, causing the surface of the liquid to behave as a thin film. Water has a high surface tension, but this characteristic does not aid in water's ability to act as a solvent.

Educational objective:

As a polar compound, water is an efficient solvent due to its ability to interact with other polar or charged compounds. The polar nature of water is due to the difference in the electronegativities of oxygen and hydrogen, along with a bent geometry. The small size of the water molecule allows efficient interactions with solutes, forming a hydration shell around and dissolving them.

Time spent: 0 sec

Subject:
General Chemistry

QID: 400475

System:
Thermodynamics, Kinetics & Gas Laws

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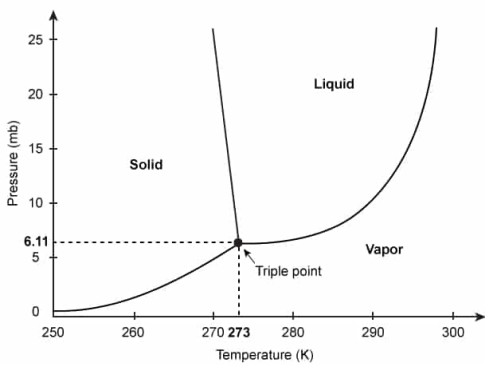


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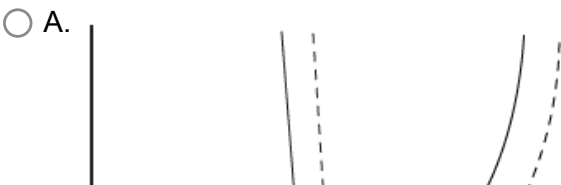
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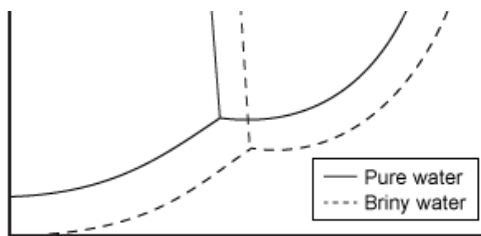
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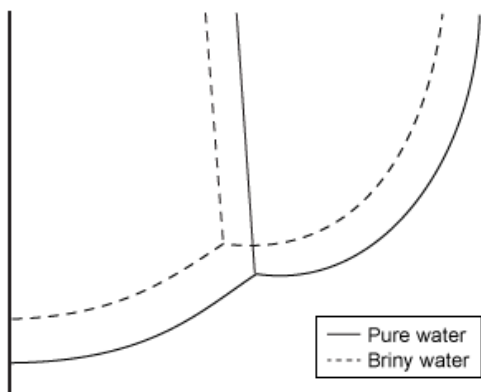
Which of the following phase diagrams best shows the change from pure water's phase boundaries to the phase boundaries of the briny water found on Mars?





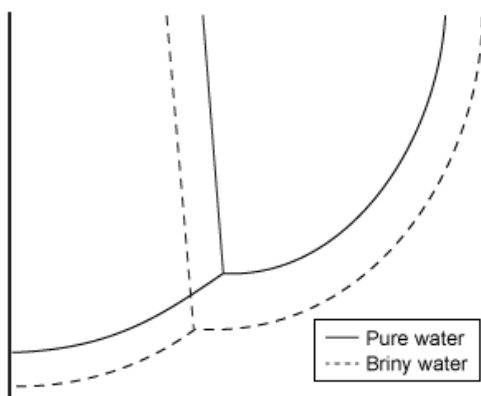
[12%]

☐ B.



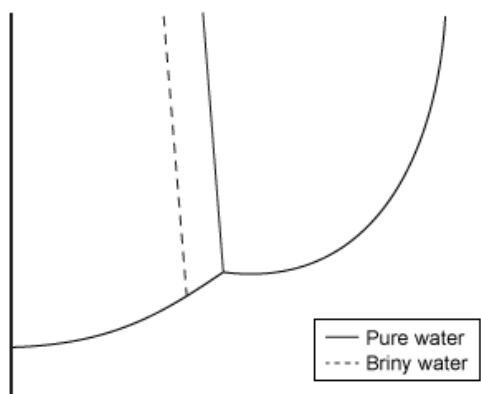
[16%]

✓ ☒ C.



[67%]

☐ D.

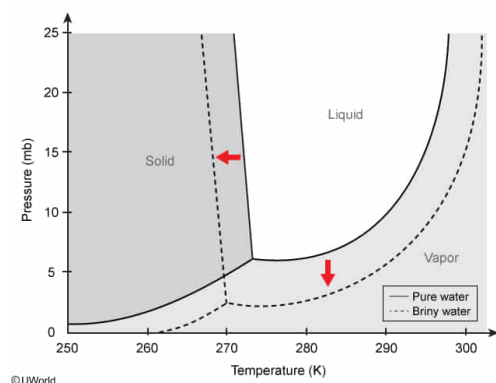


[4%]

Correct

67% answered correctly

Explanation:



A **phase diagram** shows a substance's stability in each phase as a function of **pressure (y-axis)** and **temperature (x-axis)**. The **boundary lines** between two **phases** indicate the conditions under which they are in equilibrium. According to the passage, Raoult's law states that the addition of any solute to a pure liquid **lowers** the freezing point temperature and vapor pressure of the resulting solution. A substance's phase diagram shows this change by shifting the solid-liquid phase boundary to lower temperatures and shifting the liquid-vapor phase boundary to lower pressures.

(Choice A) The addition of solute lowers the freezing point of a solution. Therefore, the liquid-solid phase boundary should shift to lower temperatures.

(Choice B) The addition of solute lowers the saturation vapor pressure of a solution. Therefore, the liquid-gas phase boundary should shift to lower pressures.

(Choice D) The addition of solute alters both the freezing point and the saturation vapor pressure. Both the solid-liquid and liquid-gas phase boundary lines should shift to lower temperatures and pressures, respectively.

Educational objective:

A phase diagram consists of phase boundary lines that illustrate the equilibrium points and transitions

between phases as a function of pressure (y-axis) and temperature (x-axis). Raoult's law states that the addition of solute to a pure substance lowers the freezing point and saturation vapor pressure of the resulting solution. On addition of a solute, the phase boundary lines on a phase diagram should move accordingly.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400476
System:
Thermodynamics, Kinetics & Gas Laws

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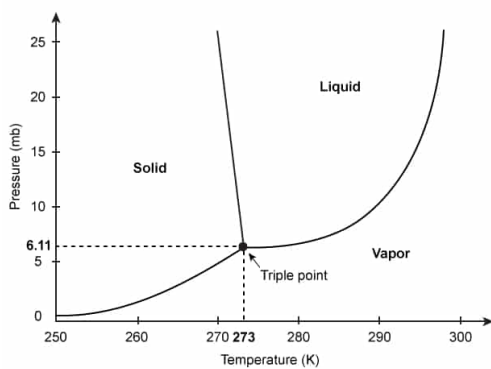


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In the liquid phase, water dissolves and transports molecules, and aids in the biochemical reactions necessary to support life. For this reason, scientists believe liquid-phase water is essential to the existence of extraterrestrial life. Although it has been known for some time that water is present in both the gas and ice phases on Mars, the question remains as to whether there are regions where liquid-phase water can stably exist. Estimates of Mars' surface temperature and pressure generated from satellite readings and global climate models, as well as observations made by Mars rovers, reveal that surface conditions on the planet can vary greatly.

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Which of the following affects the freezing point of an ideal solution consisting of a solute dissolved in water

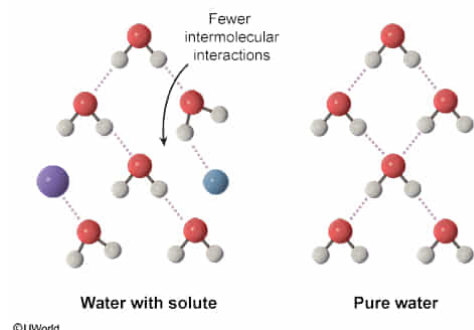
- I. The strength of intermolecular forces
- II. The amount of solute dissolved in the water
- III. The chemical identity of the solute

- ☐ A. I only [29%]
☐ B. II only [14%]
☒ C. I and II only [31%]
☐ D. I, II, and III [25%]

Correct

31% answered correctly

Explanation:



The **freezing point** is the temperature at which solid and liquid phases of a substance are in equilibrium. Freezing occurs when the **kinetic energy** of a molecule can no longer overcome the **intermolecular forces** binding it to nearby molecules. If a molecule experiences strong intermolecular forces with surrounding molecules, the kinetic energy required to overcome those forces is greater, and the **freezing point** temperature is correspondingly higher. Therefore, the freezing point of water is impacted by the **strength of intermolecular forces (Number I)**.

Raoult's law states that for an ideal solution, the addition of a solute lowers the freezing point of a liquid according to the **amount of solute** added (**Number II**). As solute is added, the solute molecules get between some of the water molecules and prevent them from interacting as easily, which interferes with the ability to freeze.

Therefore, the freezing point of an ideal solution consisting of a solute dissolved in water is affected by both the strength of intermolecular forces and the amount of solute dissolved in the solution (**Numbers I and II**).

(Number III) A dissolved solute disrupts interactions between the water molecules simply by being in between them (ie, getting in the way and preventing interactions). As such, this interference and its effect on the freezing point does not depend on the chemical identity of the solute but only on its presence. Properties that are affected by the amount but not the identity of the solute are called **colligative properties**.

Educational objective:

The freezing point of a substance is determined by the strength and extent of intermolecular forces acting between molecules. These interactions are affected by the presence of solutes. As a colligative property, the extent of freezing point depression of a solution is a function of the amount of solute added, not its chemical identity.

Time spent: 0 sec

Subject:
General Chemistry

QID: 400477

System:
Thermodynamics, Kinetics & Gas Laws

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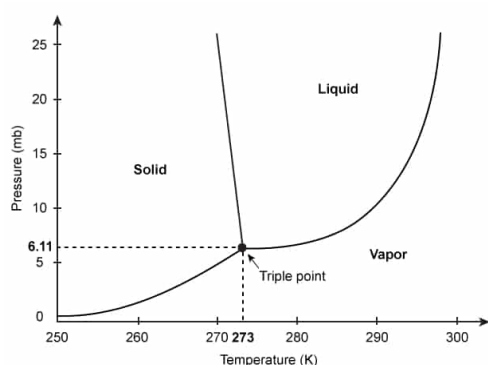


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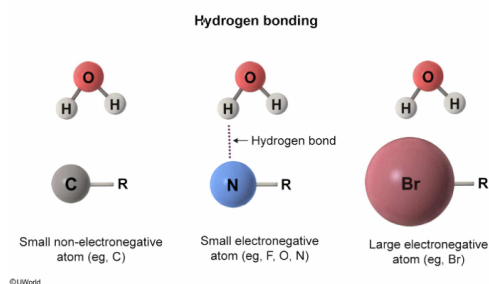
Which of the following molecules is unable to form hydrogen bonds with water?

- ✓ ☒ A. CH_3Br [93%]
- ☐ B. CH_5N
- ☐ C. $\text{C}_2\text{H}_6\text{O}_2$ [2%]
- ☐ D. HF [4%]

Correct

93% answered correctly

Explanation:



Hydrogen bonding is a strong **intermolecular force** that occurs between a **hydrogen atom** on one molecule and an **electronegative atom** with a lone electron pair on another molecule. The hydrogen atom is the bond donor and must be covalently attached to a small electronegative atom that withdraws electrons from hydrogen, giving it a partial positive charge. The partial positive charge is then attracted to the partial negative charge of the lone pair of electrons in the bond acceptor. These interactions can only occur over short distances, so they require sufficiently **small electronegative atoms** in both the donor and acceptor molecules, and are essentially limited to **fluorine, nitrogen, and oxygen**.

Carbon is a small atom but not sufficiently electronegative to participate in hydrogen bonding. On the other hand, bromine is electronegative but too large to participate in hydrogen bonding. Therefore, CH_3Br is unable to form hydrogen bonds with water.

(Choices B, C, and D) The partial positive charge on the hydrogen atom in water can form hydrogen bonds with the electronegative nitrogen, oxygen, and fluorine atoms in CH_5N , $\text{C}_2\text{H}_6\text{O}_2$, and HF , respectively.

Educational objective:

Hydrogen bonding occurs when a partially positive hydrogen atom in one molecule is attracted to the partial negative charge on an electronegative atom in another molecule. Atoms involved in hydrogen bonding must have small atomic radii, and are essentially limited to fluorine, oxygen, and nitrogen.

Time spent: 0 sec

Subject:
General Chemistry

QID: 400478

System:
Thermodynamics, Kinetics & Gas Laws

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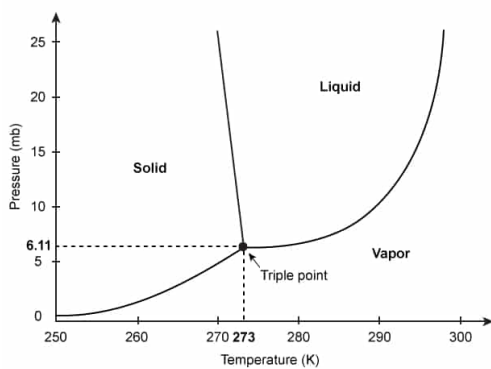


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Compared to the boiling point of a sample of pure water on Mars, the boiling point of a sample of briny water under the same atmospheric conditions would be:

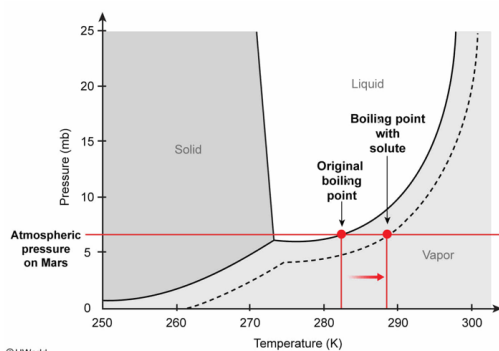
- ☐ A. lower, because the vapor pressure of briny water is higher than the vapor pressure of pure water. [14%]
- ☒ B. higher, because the vapor pressure of briny water is lower than the vapor pressure of pure water. [42%]

- ☐ C. lower, because the vapor pressure of briny water is lower than the vapor pressure of pure water. [36%]
- ☐ D. higher, because the vapor pressure of briny water is higher than the vapor pressure of pure water. [6%]

Correct

42% answered correctly

Explanation:



The vapor pressure of a substance is the partial pressure of the gas that is released from the liquid phase at any given temperature. On a phase diagram, the vapor pressure is shown as the boundary line between the gas and liquid phases, and it increases with temperature. When the temperature is high enough for the **vapor pressure** to equal the **ambient pressure**, the liquid boils.

Ambient atmospheric pressure is not affected by whether or not a liquid contains solutes, so the liquid must reach the same vapor pressure to boil regardless of the presence or absence of solute. As such, the vapor pressure achieved by the briny water at its boiling point will be the same as the vapor pressure achieved by a sample of pure water at its boiling point. The difference will be in the temperature of the two boiling points.

According to **Raoult's law**, the vapor pressure of a liquid decreases (dashed line) at any given temperature when solute is added. Briny water will therefore need to be heated to a **higher temperature** for its vapor pressure to reach the same ambient pressure and begin to boil. Therefore, the **boiling temperature** (boiling point) of briny water is higher than that of pure water under the same atmospheric conditions.

(Choices A, C and D) Because ambient atmospheric pressure is the same for both pure water and briny water, each solution must reach the same vapor pressure (equal to ambient pressure) in order to

boil. The added solute in the briny water will *lower* its vapor pressure at any given temperature (Raoult's law), causing the briny water to require a *higher* temperature to raise its vapor pressure enough for the solution to boil.

Educational objective:

The boiling point temperature is the temperature at which the vapor pressure is equal to ambient pressure. The addition of solute lowers the vapor pressure of a solution at all temperatures, and therefore raises the temperature required for the vapor pressure to become equal ambient pressure and begin to boil (the boiling point).

Time spent: 0 sec

Subject:
General Chemistry

QID: 400479

System:
Thermodynamics, Kinetics & Gas Laws